Sanjit Dinesh Krishnan

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Professional Summary:

Sanjit Dinesh Krishnan is a Computer Science undergraduate at R.V University, Bangalore, specializing in **AI/ML and Robotics**. His academic trajectory reflects a purposeful progression from physical systems to artificial intelligence. Grounded in hands-on hardware experience — from wiring Arduino sensors for IoT systems to programming PLC ladder logic for industrial automation — he built an intuitive understanding of how machines perceive and interact with the real world. This strong foundation in embedded systems naturally led him to pursue a deeper question: *how can these machines learn to think?*

The answer emerged through his intensive work in **deep learning and computer vision**, where he built a CNN-based document forgery detection system using transfer learning with ResNet-18, independently designing the complete ML pipeline from synthetic dataset generation to model evaluation. Simultaneously, he explored **robot kinematics** by implementing differential drive simulations in Python, bridging the gap between software intelligence and physical motion.

His internship at the **Indian Institute of Astrophysics** tested these converging skills under real institutional pressure. Tasked with building a complex multi-role web platform within a single month, he delivered a production-grade system featuring real-time auto-save, PostgreSQL-backed workflows, and a robust authentication architecture — demonstrating his ability to ship reliable software when the stakes are high. Sanjit is now seeking his next student internship to deepen his expertise at the intersection of **AI, machine learning, and robotics** — contributing to teams building technology that senses, reasons, and acts.

Education:

Degree / Board	Institution	Year	Score
B.Tech in Computer Science (Minor: Robotics & Industrial Automation): 4 Years	R.V University, Bangalore	2024 – Present	Sem 1: 8.0 → Sem 2: 8.5 → Sem 3: 8.5 GPA
High School, CBSE Board	The Amaatra Academy, Bangalore	2024	7.0 CGPA
Matriculation, ICSE Board	Notre Dame Academy, Bangalore	2022	9.4 CGPA

Internship Experience:

Web Application Development Intern — Indian Institute of Astrophysics

Summer 2025 (Post 2nd Semester) — Bangalore, India

The Indian Institute of Astrophysics required a complex, multi-role digital ecosystem to manage its internal workflows — a system where Applicants, Chairs, and Referees each operated within distinct permission boundaries, yet shared a seamless, unified interface. Long-form applications risked data loss, email verification systems needed to be bulletproof, and the platform had to be delivered within an aggressive one-month timeline.

What I Built: Developed a Python-based web platform featuring a **dynamic "Auto-Save" form system** that eliminated the traditional submit-and-reload pattern. The frontend was engineered to synchronize instantly with a **PostgreSQL database**, transforming a tedious multi-page application process into a fluid, single-page experience where no data was ever lost mid-session.

The Hard Problem I Solved: The project stalled when the account activation system failed to reliably verify emails. SMTP's handshakes were unreliable, and token-based link expiration introduced edge cases that broke the user flow. Rather than accepting a workaround, Sanjit debugged the authentication pipeline end-to-end, resolved the SMTP integration issue, and implemented a robust token verification mechanism.

Outcome: Within one month, delivered two critical modules — the core applicant-to-chair workflow and the authentication system — providing a functional foundation that met institutional deadlines.

Dimension	Detail
Tech Stack	Python, PostgreSQL, Bootstrap, HTML, JavaScript
Key Challenge	Multi-role authentication with SMTP email verification
Delivery	2 critical modules shipped in 1 month under institutional deadline
Impact	Production-grade platform serving IIA's internal review process

Technical Projects:

AI and Computer Vision

Fake ID Identification System — *3rd Semester College Project*

Built a CNN-based document forgery detection system using **transfer learning with ResNet-18** to classify synthetically generated ID images as genuine or forged. The project spanned the full machine learning pipeline: researching CNN architectures, generating a synthetic dataset with controlled forgeries, implementing fine-tuning with data augmentation, training with Cross-Entropy loss and Adam optimizer, and evaluating model performance using precision, recall, and F1-score metrics.

Skills Applied: Deep Learning, Transfer Learning, Computer Vision, Image Preprocessing, Data Augmentation, Model Fine-Tuning, End-to-End ML Pipeline Development

Robotics and IoT

IoT-based Smart Car Parking System — *3rd Semester College Project*

Developed an IoT-based Smart Car Parking System using **Arduino Uno and ESP32** to monitor parking space availability and enable automated parking management. Led the hardware setup and system integration, including circuit design, sensor connections, and microcontroller interfacing — building a real-time embedded system from the ground up.

Skills Applied: IoT Hardware Design, Arduino & ESP32 Programming, Sensor Integration, Real-Time Embedded Systems

Differential Drive Robot Simulation — *Robotics Coursework*

Implemented a simulation of a **differential drive robot** using Python and Pygame, applying both forward and inverse kinematics to achieve autonomous motion behaviors. This project bridged theoretical robotics with practical software simulation, demonstrating understanding of how physical robots translate motor commands into coordinated movement.

Skills Applied: Robot Kinematics (Forward & Inverse), Python, Pygame, Autonomous Motion Planning

Full-Stack Development

Car Rental Management System — *2nd Semester College Project*

Independently built an end-to-end Car Rental Management System with user authentication, car listings, bookings, and full CRUD database operations. Developed the complete frontend, backend, and MySQL database using PHP, HTML, and CSS — gaining foundational experience in relational database design and full-stack architecture.

Skills Applied: PHP, MySQL, HTML/CSS, Relational Database Design, CRUD Operations, Full-Stack Development

Technical Skills:

Category	Technologies
Programming Languages	Python, JavaScript, PHP, HTML/CSS, SQL
AI / Machine Learning	Deep Learning (CNNs), Transfer Learning (ResNet-18), Computer Vision, Data Augmentation, Model Fine-Tuning
Robotics & Embedded	Arduino, ESP32, PLC Ladder Diagram Programming, Differential Drive Kinematics, IoT Sensor Integration
Web Development	PostgreSQL, MySQL, Bootstrap, Full-Stack (Frontend + Backend + Database)
Tools & Platforms	Git/GitHub, Arduino IDE, PLC Editor Software, Pygame

Leadership and Extracurricular Achievements:

- As **Sports Captain** at Notre Dame Academy, Sanjit led teams across inter-school tournaments while competing at the **district level and Indian League (Under-14) in football**, earning Best Player awards in multiple open tournaments.
- He represented both Notre Dame Academy and Amaatra Academy as **School Captain** in various inter-school competitions.
- In the arts, he represented Notre Dame Academy in **multiple inter-school singing competitions**, winning awards that reflect his creative versatility alongside his technical pursuits.

Seeking a student internship in AI/ML, Robotics, or Intelligent Systems for Summer 2026. References available upon request.